

QK2 Standards Register

Every BIP/SLIP/RFC the QuietKey QK2 co-signer touches, with a candidate QK2-*adoption* ruling — plus the native-P2WSH derivation path worked into a checksum-valid reference descriptor. A ruling here is a proposed adoption, not a claim about the standard's own status or an implementation.

READ THE FOUR STATUSES SEPARATELY

Each entry has (1) the **standard's own status** (Deployed / Draft / Withdrawn), (2) QK2 **adoption** (in/out/deferred), (3) **implementation** status (none yet), and (4) **test** status (coordinator interop pending). These are not the same thing. The wallet is **SELECTED product direction**: native-P2WSH `wsh(sortedmulti(2, A, B, C))` over BIP-48 — the *policy* (2-of-3, independent legs A/B/C) is settled. "SELECTED" is a chosen product direction, **not** an implementation or interop result: the descriptor remains subject to the coordinator-interoperability matrix, and no single descriptor is validated on target hardware yet.

WALLET	ACCOUNT PATH	KEY ORDER	PSBT	TAPROOT	NETWORK
wsh 2-of-3 (sel.)	m/48'/0'/0'/2'	sortedmulti	v0 (v2 deferred)	out (sign)	mainnet only

1. [The one path that matters](#)

2. [The standards register](#)

3. [Recipient address rendering](#)
4. [Scope-boundary rulings](#)

5. [Internal formats to version](#)

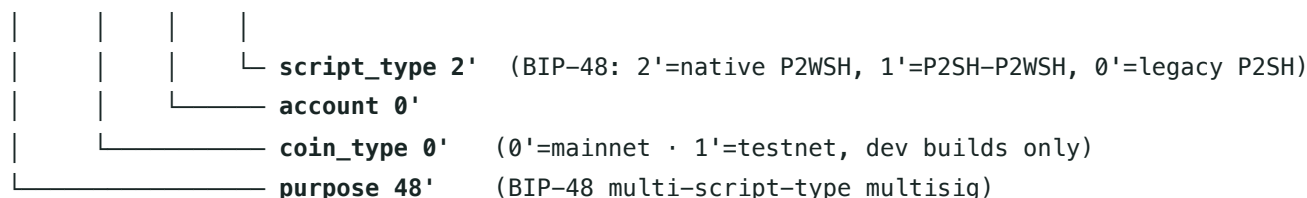
6. [The miniscript escape hatch](#)

The exact derivation path, worked end-to-end

A wrong path or a wrong key order sends change to an address the device can never reach again.

This is the one thing that must be unambiguous, so it is spelled out and then computed into a candidate reference descriptor and address. Loadability into Sparrow or ColdCard is an **interoperability objective pending repeatable cross-coordinator tests**, not a confirmed result.

ACCOUNT PATH m / 48' / 0' / 0' / 2'



LEAF .../2' / {0,1} / i 0 = receive branch, 1 = change branch, i = index (both NON-hardened)
(' and h both denote hardened; the descriptor below emits h)

REFERENCE WORKED EXAMPLE (REGENERATED, CANDIDATE)

A reference computation built from two candidate QKEC-1 seeds (leg A = dice vector TV-A2, leg B = hybrid vector TV-B2). The vector chain was **regenerated at 100 dice rolls** (conforming to the 24-word/≥100-roll rule) and cross-checked against the published BIP-32 and bech32 reference vectors and secp256k1 73c5da0a — evidence level 2 (independent cross-check), not a target-hardware result. The leg fingerprints are a reference computation, not canonical values: the entropy → key → address chain is internally consistent. The **policy is now settled** (native-P2WSH 2-of-3); what remains open is the Seed-A protection decision and the coordinator-interoperability matrix.

-- SELECTED policy: native P2WSH 2-of-3 (independent legs A/B/C). This is the primary --
-- worked example. Policy is settled; the descriptor is subject to interop tests, not --

-- validated on target hardware. The 2-of-2 below is a HISTORICAL/REGRESSION fixture only. --

SELECTED (primary) native P2WSH 2-of-3 (independent legs A/B/C), two ordinary descriptors as authority (BIP-389 multipath optional), BIP-wsh(sortedmulti(2,
[2fae9711/48h/0h/0h/2h]xpub6E4Ac9VDE5KxfwbTXAkZacVZ3XmbMo32suyVgt2mgn8mAZbyUuNEDXuZu9D396UYikuXXJ4tFkxE89fHNboRyNzfMzG2sph4V2MuourfoYr/
[72a14ab8/48h/0h/0h/2h]xpub6ETbdARx7dNLzy6RiMBLWHkr39b2KhiWsu1h3RQBp5DT79KCU5b5vhL2RfbEmVf4aW6yi8hQPkY8cNxBpjAS7USKboAguDQR8CRPgctyuff/
[badaa8ff/48h/0h/0h/2h]xpub6EPnBYhH8bEknocsxAw8VtsVYzMT1sBgbUFSpwikQMCMYhV1tQ38TbhrHuDVlbeeodE99AA9x8LEHjLsHsMHQg36BGBLsqd3q3CrWq1tQ1U/
))#kuyqfyj3
first receive address m/48'/0'/0'/2'/0/0 bc1qecahkw0srs8xpmgwqya49qtknxwkwrt0ky4jds52scsw4etmd6s5s93pr

HISTORICAL / REGRESSION FIXTURE native P2WSH 2-of-2 (legs A/B) – NOT the product wallet; retained only as a regression vector
wsh(sortedmulti(2,
[2fae9711/48h/0h/0h/2h]xpub6E4Ac9VDE5KxfwbTXAkZacVZ3XmbMo32suyVgt2mgn8mAZbyUuNEDXuZu9D396UYikuXXJ4tFkxE89fHNboRyNzfMzG2sph4V2MuourfoYr/
[72a14ab8/48h/0h/0h/2h]xpub6ETbdARx7dNLzy6RiMBLWHkr39b2KhiWsu1h3RQBp5DT79KCU5b5vhL2RfbEmVf4aW6yi8hQPkY8cNxBpjAS7USKboAguDQR8CRPgctyuff/
))#w9nvs8vx
first receive address m/48'/0'/0'/2'/0/0 bc1qfuszf9fjyp5wshvcjazjpy3xj7dt7mec44lc5pz380rulst5yzqq9yjge
first change address m/48'/0'/0'/2'/1/0 bc1q4ufuyqar8z2r9vafyrzlg2w7usfeeqzw56f28ammzdzl2m3ykuhsc25e58

Per-key sorted pubkeys (BIP-67) and witnessScript bytes are regenerated in the fixtures – they change with the new xpubs; BIP-67 lexicographic ordering still applies at expansion time.

Why each choice

- **BIP-48, script_type 2'** is the settled convention for native-segwit multisig across ColdCard, Sparrow, Nunchuk, and Specter. Do not invent a path; interop depends on this exact one.
- **sortedmulti (BIP-67 lexicographic ordering)** makes cosigner order irrelevant — the xpubs can be registered in any order and every wallet derives the same scripts. This removes an entire class of "cosigner order mismatch" footguns that plain multi() carries.
- **Two ordinary checksummed descriptors are the conservative recovery authority** — a receive /0/* and a change /1/*, each a plain single-branch wsh(sortedmulti(2, ...)) that any BIP-380 coordinator can import. This is the decades-long authoritative representation of the wallet.
/<0;1>/* **multipath (BIP-389, still Draft)** is an *optional convenience only* — one registered

descriptor covering both branches, expanded on-device — and is **never the sole recovery record**. Verification must cover both branches (SeedSigner's `/0/*` → `/0,1/*` rewrite exists for exactly this); test every supported coordinator's multipath handling before relying on it anywhere.

- **The 8-char BIP-380 checksum** (`#kuyqfyj3` for the selected 2-of-3; `#w9nvs8vx` is the retired 2-of-2 regression fixture) is shown to the user at registration as an anti-xpub-swap *error-detection* anchor — a single flipped character in any xpub changes it completely. **A checksum is error-detection, not cryptographic authentication:** an attacker can build a *different* valid descriptor carrying its own valid checksum. Authentication therefore comes from the provisioning **derive+compare** step (next bullet), not from the checksum.
- **Descriptor authenticity = derive+compare at provisioning.** At provisioning the terminal derives every A/B/C origin/xpub from the *actual* factors and compares them to the descriptor, shows a compact policy identity plus the first receive/change addresses, and requires a scan-back / re-import rehearsal. During recovery, D-derived scripts must match the funded UTXOs. The checksum alone never establishes that the descriptor describes *your* keys.

02 THE REGISTER

Every standard, ruled

The four axes are kept in separate columns: **STD STATUS** = the standard's own maturity (Deployed / Draft / Withdrawn) · **QK2 ADOPT** = the proposed adoption ruling · **IMPL** · **TEST** = build and coordinator-interop status. Nothing here is implemented or interop-tested yet, so IMPL is uniformly *none* and TEST *pending*.

Adoption legend: **IN** proposed as normative for v1 (not yet built) · **DISPLAY** parse/render recipient data only, never generate · **RECOMMEND** candidate · **DEFER** documented but deferred · **OUT** excluded, and why.

STANDARD	STD STATUS	QK2 ADOPT	IMPL · TEST	WHAT IT IS / WHY THE RULING
A · KEY DERIVATION & SEED				
BIP-32	Deployed	IN	none · pending	HD derivation. Enforce non-hardened index bounds $0 \leq i < 2^{31}$; unit-test the $i == 2^{31}$ boundary (iancoleman has a known off-by-one there).
BIP-39	Deployed	IN	none · pending	Mnemonic = Seed A representation and the Mode A dice encoding. English wordlist only; PBKDF2 fixed at 2048 rounds; NFKD on both inputs.
BIP-39 passphrase	Deployed	OUT	—	QK2-04 excludes it — hardcode empty. The second factor is the smart card + Recovery Document, not a "25th word".
BIP-43	Deployed	IN	none · pending	Purpose-field framework underneath BIP-48.
BIP-44	Deployed	PARTIAL	none · pending	The <code>purpose'/coin'/account'</code> skeleton is reused via BIP-48; BIP-44 P2PKH single-sig addresses are not generated (recipient P2PKH still rendered — see §03).
BIP-45	Deployed	OUT	—	Older cosigner-indexed P2SH multisig; superseded by BIP-48 for modern P2WSH.
BIP-48	Deployed	IN ★	none · pending	Defines <code>m/48'/. . . /2'</code> for native-P2WSH multisig — the SELECTED QK2 convention (subject to interoperability, not policy selection); coordinator interop tests still pending, not an infallible recovery mechanism. All three key origins A/B/C use the same convention.
BIP-49 / 84 / 86	Deployed	OUT	—	Single-sig nested / native segwit / Taproot. QK2 is multisig-only; 86 also excluded by the Taproot ruling.
BIP-85	Draft	OUT (v1)	—	Deterministic child entropy. The mechanism <i>if</i> duress/child wallets are later added; not in the single-sealed-wallet v1.

B · DESCRIPTORS (BIP-380 FAMILY)				
BIP-380	Deployed	IN	none · pending	Output descriptors + the 8-char checksum. Show the checksum at registration for user cross-check.
BIP-382	Deployed	IN	none · pending	wsh(...) — the wallet's script wrapper.
BIP-383	Deployed	IN	none · pending	sortedmulti(2, ...) — multisig with BIP-67 ordering (SELECTED 2-of-3, independent legs A/B/C).
BIP-389	Draft	OPTIONAL	none · pending	Multipath /<0;1>/* — one descriptor, both branches. Optional Draft convenience only, never the sole authority. The conservative recovery representation is two ordinary single-branch checksummed descriptors (/0/* and /1/*), which any BIP-380 coordinator can import; test every supported coordinator before relying on multipath anywhere.
BIP-388	Complete	RECOMMEND	none · pending	Wallet policies — the HW-signer registration format designed for limited displays; candidate on-device descriptor representation.
BIP-67	Deployed	IN	none · pending	Lexicographic public-key ordering used by sortedmulti.
BIP-381	Deployed	DISPLAY	none · pending	pk/pkh/sh — recognise legacy recipient scripts for correct display only.
BIP-384/385/386/387	Deployed	OUT	—	combo, raw/addr, and the Taproot descriptors tr / multi_a — not generated (386/387 excluded by Taproot ruling).
C · PSBT				
BIP-174	Deployed	IN	none · pending	PSBT v0 — the core signing container (adopted v0 profile). microSD carries binary PSBT v0 only — no base64/hex autodetection and output encoding is not inferred from

				...recovery, and output encoding is not influenced from the input.
BIP-370	Deployed	DEFER	—	PSBT v2. Deferred pending a fresh review; the v0/v2 relationship is not treated as an operative exclusivity rule. Harden the PSBT v0 profile first.
BIP-371	Deployed	OUT	—	Taproot PSBT fields — excluded by the Taproot ruling.
D · TRANSACTION, SCRIPT & ADDRESS ENCODING				
BIP-141 / 144	Deployed	IN	none · pending	Segwit structure and witness serialization.
BIP-143	Deployed	IN ★	none · pending	Segwit v0 sighash. This is where the input-amount commitment lives — the CVE-2020-14199 defences attach here. QK2 accepts only SIGHASH_ALL in v1.
BIP-173	Deployed	IN	none · pending	Bech32 — our P2WSH v0 addresses. (The Recovery Document footer is a <i>separate</i> custom base32 with a Bech32-derived polymod, not this BIP — see §05.)
BIP-350	Deployed	DISPLAY	none · pending	Bech32m — never generated by us, but recipient Taproot/v1+ addresses must render correctly so the user sees the true destination.
BIP-13 / 16	Deployed	DISPLAY	none · pending	P2SH address + script — recognise P2SH-class recipients.
BIP-125	Deployed	DISPLAY	none · pending	RBF signalling — show whether the transaction is replaceable.
BIP-65 / 68 / 112	Deployed	OUT (v1)	—	CLTV / relative-locktime / CSV — only relevant if the miniscript recovery branch is adopted (§06). The sketched older(N) branch is a relative timelock and pulls in BIP-68 + BIP-112 only . BIP-65 (CLTV) is for an ABSOLUTE after() locktime and does NOT apply to a relative older(N) .

E · SIGNATURES				
RFC 6979	RFC	IN	none · pending	Deterministic ECDSA nonces. The terminal uses exact deterministic RFC 6979 (pinned libsecp256k1, strict DER, low-S, verify-before-release) — not a "hedged"/TRNG-mixed <code>k</code> , which is a different construction that would need an exact reviewed algorithm plus card support. Card signature behaviour is a Gate-B capability to specify and test.
low-S	Policy	IN	none · pending	Low-S signatures, sourced independently. BIP-146 is withdrawn and must not be cited as a live normative dependency; specify low-S through the actual library and current Bitcoin validation/standardness policy instead.
low-R grinding	Policy	OPTIONAL	—	A signature-size optimization (uniform 71-byte sigs), not "canonical" and not required for correctness.
Anti-exfil (s2c)	Experimental	DEFER	—	Sign-to-contract (secp256k1-zkp <code>ecdsa_s2c</code>) — the upstream module is experimental with an unstable API , and one-shot JavaCard ECDSA likely cannot implement it. Optional future hardening, not a locked or mandatory primitive; deferred pending card feasibility (Card Protocol §02).
BIP-340	Deployed	OUT	—	Schnorr — excluded with Taproot; also gated on whether the JavaCard can do Schnorr at all.
F · SLIP, BACKUP & TRANSPORT				
SLIP-132	SLIP	REJECT	—	ypub/zpub version bytes conflate script type with key — reject in descriptors; require xpub + explicit script type. Convert only for display.
SLIP-39	SLIP	OUT	—	Shamir backup — a different scheme from QK2's recovery

				record. QK2's Recovery Document (A1) carries authenticated ciphertext of 256-bit Seed-A entropy; encryption is MANDATORY (decryption key = A2, duplicated on cards B and C). The complete descriptor/policy record (D) is privacy-sensitive, not public — kept offline + encrypted at rest, redundantly across the custody objects, never "copied widely".
BBQr (PSBT framing)	Vendor	IN	none · pending	BBQr (file type P , uncompressed base32) is the SELECTED v1 QR PSBT framing. The reviewed SeedSigner and COLDCARD both support it; its sequential part model is simpler to bound than a fountain decoder, and uncompressed mode avoids a decompression-bomb surface. Both QR and microSD are REQUIRED transports (microSD is the lower-risk first path). BC-UR 2.0, Specter fragments, and SeedQR are NOT PSBT-framing candidates — do not promise simultaneous formats or generic QR autodetection. The FORMAT is selected, but QR capture/animated-reassembly/screen-export/RAM/performance on the existing enclosure/optics/320×240 display remains a HW/system validation BLOCKER (Gate A) — no mechanical redesign is authorized, and hardware performance is not yet proven. Camera has two mode-locked uses only — A1 footer capture and PSBT QR capture — and never auto-detects a seed QR.
SeedQR / Compact	Vendor	OUT	—	No seed-word QR import. The camera has two mode-locked uses only — A1 footer capture and PSBT QR capture — and never auto-detects a seed QR or accepts arbitrary QR classes. Disabling seed import shrinks the attack surface (a mis-classified QR can't inject a seed). Seed-A entropy lives as <i>authenticated ciphertext</i> in the A1 footer, not as naked BIP-39 words.

BIP-329	Draft	OUT (v1)	—	Label export — deferred; interacts with statelessness.
BIP-322	Complete	OUT (v1)	—	Generic message signing — a Complete standard, but the adoption ruling still keeps it OUT of v1: it adds parser surface for no fund-safety gain in v1.
BIP-21	Deployed	OUT	—	Payment URIs — the device has no companion/URI intake.
BIP-379 (miniscript)	Draft	OUT (v1)	—	Draft status. Not needed for standard 2-of-3. Only a timelock recovery option would pull it in, adding a Draft-spec satisfaction path and its audit burden to the trusted signer (§06) — a rejected alternative to standard multisig.
G · QUANTUM EXPOSURE & PQ READINESS — QUANTUM-PREPARED, NOT QUANTUM-PROOF				
P2WSH (vault output)	Deployed	IN ★	none · pending	Quantum exposure: good long-exposure — the on-chain output commits only to SHA256 of the witness script, so dormant keys are hash-hidden. But it REVEALS the script and EC public keys when spent (short mempool window). Mitigate with no address reuse and, in Quantum Shelter mode, broadcast a finalized raw transaction (no PSBT xpub/origin metadata back online).
P2TR (Taproot)	Deployed	OUT (vault)	—	Quantum exposure: a P2TR output exposes an EC output key immediately (before any spend). Excluded for the vault for exactly this long-exposure reason, on top of the existing Taproot ruling. Recipient P2TR is still rendered for display (§03, BIP-350).
xpubs / descriptors	Deployed	PRIVACY-SENSITIVE	none · pending	Quantum exposure: xpubs reveal EC public keys (BIP-360 warns), and a descriptor exposes the whole policy. D is not "public, copy widely" — keep it offline + encrypted at rest, no cloud sync, no account-xpub/full-descriptor export to the online coordinator by default; export

					individual P2WSH receive addresses/packs for monitoring only.
BIP-360 / P2MR	Draft	MONITOR / DEFER	—		Draft. Pay-to-Merkle-Root / quantum-resistant output proposal. Gives no short-exposure protection without PQ signatures , so not adopted ; monitored/deferred pending a credible reviewed PQ output/signature path. Migration trigger: sweep from hidden-key P2WSH outputs when such a path deploys or CRQC progress changes the horizon.
NIST ML-DSA / SLH-DSA	FIPS 204/205	OUT (not consensus)	—		PQ signature standards exist but are NOT Bitcoin consensus . Do not attach an off-chain PQ signature to a PSBT and market it as protection, and do not freeze a proprietary seed-to-PQ derivation now. Reserve versioning + benchmark larger PQ payloads; migrate to future PQ Bitcoin standards when they land.
H · RECOVERY-DOCUMENT & ENTROPY CRYPTO (RFC)					
RFC 8439	RFC	IN ★	none · pending		ChaCha20-Poly1305 (IETF AEAD) — the SELECTED A1 capsule AEAD (§05). Portable and constant-time on the ARM target without relying on hardware AES; fresh stored 96-bit nonce, 128-bit tag, no plaintext released before the tag verifies . This settles the A1 crypto layer; only RS/font/OCR print constants stay behind Gate A.
RFC 5869	RFC	IN	none · pending		HKDF-SHA256 — the QKEC-1 entropy conditioner (Extract/Expand, per-purpose Seed-A / Signer-B / Signer-C / A2 , each purpose independent) and the A1 document key HKDF-SHA256(IKM=A2, salt=wallet_id, info="QuietKey/A1/v1") .

★ = highest-consequence: getting these wrong loses funds or breaks the security model.

Recipient address types the device must render – even the ones it never makes

QK2 signs only for its own P2WSH wallet, but it *pays* to arbitrary destinations. The device screen is the only trusted surface, so it must decode and render **every** standard output type at full length. Mis-rendering a Taproot (bech32m) recipient as garbage — or worse, as a valid-looking wrong address — is an address-substitution vector.

OUTPUT TYPE	PREFIX	ENCODING	RULING
P2PKH (legacy)	1...	base58check	RENDER
P2SH / nested segwit	3...	base58check	RENDER
P2WPKH	bc1q... (20-byte)	bech32 (BIP-173)	RENDER
P2WSH	bc1q... (32-byte)	bech32 (BIP-173)	RENDER + own
P2TR (Taproot)	bc1p...	bech32m (BIP-350)	RENDER
OP_RETURN	–	raw push	RENDER raw
bare / unknown script	–	–	FATAL (v1)

RENDERING RULES

Full address, no truncation, chunked in 4-char groups over two lines, scroll-gated so the approve control appears only after the whole address is seen. Parse OP_RETURN push opcodes correctly (direct push vs OP_PUSHDATA1/2 — never a fixed data[3:] slice). An

unknown scriptPubKey is never silently shown as "0 BTC to unknown" — an unknown or unrenderable output script is **fatal in v1** (the transaction is rejected, not merely warned on), matching the PSBT rule.

04 SCOPE BOUNDARIES

Rulings that bound the whole surface

QUESTION	RULING	RATIONALE
Network	mainnet only (coin 0')	Production firmware is mainnet-only. Testnet (coin 1') lives solely in a separate dev/simulator build — avoids the ColdCard 3.2.1 testnet-address-confusion social-engineering class on a real device.
Wallets held	native P2WSH 2-of-3 (SELECTED)	One wallet policy: <code>wsh(sortedmulti(2, A, B, C))</code> with independent legs A/B/C. The threshold is settled product direction , not an open decision. Multiple accounts/descriptors are OUT for v1; each added wallet multiplies the registration, address-explorer, and change-verification surface.
Spending control	2-of-3 threshold (no PIN)	Not a BIP-39 passphrase and NO card PIN / OAC / duress / retry-counter / lockout . B and C are bearer objects — physical custody <i>is</i> authorization; the 2-of-3 policy is the spending control. This matters most for C: an inheritance kit must work after years with no forgotten PIN, vendor, account, or clock.
Cosigner order	sortedmulti (BIP-67)	Order-independent, so a mis-ordered registration can't silently produce a different (unrecoverable) wallet.
Address type generated	native P2WSH only	No legacy/nested/Taproot generation. One script type to derive, verify, and test.

Non-BIP formats that still need version bytes & agility

These are QuietKey-specific and long-lived, so each needs an explicit version byte and algorithm-agility plan, versioned as visibly as any BIP.

- **QKEC-1** — the entropy conditioner (**HKDF-SHA256** Extract/Expand, per-purpose Seed-A / Signer-B / Signer-C / A2, each purpose independent). **CANDIDATE** (version byte 0x01; reference spec + regenerated vectors at evidence level ≤ 2 — not frozen).
- **Recovery Document capsule (A1 footer)** — the AEAD-protected recovery structure (draft wire version 0x02). **Encryption is MANDATORY.** The crypto layer is now **SETTLED**: AEAD = **ChaCha20-Poly1305 (RFC 8439, IETF construction)** from a reviewed constant-time implementation — portable and constant-time on the ARM target without relying on hardware AES. Document key = `HKDF-SHA256(IKM=A2, salt=wallet_id, info="QuietKey/A1/v1")`; **fresh random 96-bit nonce, stored in the capsule** (no implicit nonce); 128-bit tag; **no plaintext released before the tag verifies**. Plaintext is exactly 32 bytes of Seed-A entropy; AAD binds magic, coding version, crypto version, network, and the full `wallet_id`. Compact payload: `version fields | nonce | ciphertext(32) | tag(16)`. The old 49/83/133/142 example assumed a 49-byte capsule + implicit nonce and is **NOT normative**. The permanent document format is **not** coupled to whatever AEAD one JavaCard happens to expose: the bearer card releases **A2** (the decryption key, exportable by design) over a minimal card session, and the *replaceable terminal* performs the AEAD decryption. Only the **RS/font/OCR/footer-length print constants remain unfrozen behind Gate A**; recompute the print vectors once the codec length settles.
- **QKRD-1 recovery encoding** — the OCR-carried recovery footer. It is a **custom base32 format, not Bech32**; its checksum is a *Bech32-derived polymod*, not a BIP-173 / BIP-173-checksum. Character count, RS parameters, and grouping are **unfrozen** and pending the Seed-A protection

decision (e.g. a random 12-byte nonce alone pushes the capsule past 60 bytes, which would move the RS parameters and the footer length off any fixed figure). Include a version field and a human-readable fallback for permanent OCR failure.

- **Complete descriptor / recovery-metadata survival (D)** **STOP-SHIP** — a **normative backup requirement, not merely an interop example**. A 2-of-3 output cannot be spent in recovery without the **complete recovery metadata** surviving **independently of every card, terminal, and coordinator: two ordinary checksummed receive/change descriptors** as the conservative authority, all A/B/C xpubs + fingerprints + origins, threshold/script-type/network/derivation, policy version, birthday, first receive/change addresses, and recovery/rotation instructions. D is **privacy-sensitive, NOT "public, copy widely"** — its xpubs expose EC public keys (BIP-360 warns). Posture: **offline + encrypted at rest, no cloud sync**; store on B and redundantly in the C kit (Simple/Inheritance), or as a separate cloaked/encrypted "wallet map" (Quantum Shelter). **Authenticity = derive+compare**: derive every A/B/C origin/xpub from the actual factors at provisioning and compare — a checksum is error-detection, **not** authentication.
- **Registered-descriptor sealing** — the on-card representation (Gate B, bench-required). Recommend BIP-388 wallet policies as the candidate wire form; version it. This on-card copy does **not** satisfy the independent-survival requirement above — the complete descriptor must also live somewhere no single device controls.

06 FORWARD HOOK

The one place miniscript comes back – and why it matters

REJECTED ALTERNATIVE – KEPT ONLY AS A SHORT APPENDIX

The **SELECTED** wallet is native `wsh(sortedmulti(2, A, B, C))` — recovery lives entirely inside ordinary `sortedmulti` with no new standards (B lost → A1+C; A1 lost → B+C; C lost → A1+B then rotate C). A miniscript timelock — spend A+B normally, or A alone after

a relative timelock — is a **REJECTED alternative** retained only as an option-study sketch, alongside cloned-B, encrypted-B-export, and single-sig (all rejected). Sketch:

```
wsh( or_d( multi(2, A, B),  
            and_v( v:older(<N>), pk(A) ) ) ) ; A+B anytime; A alone after N blocks (~1yr)
```

Choosing the timelock would flip **BIP-379 (Draft miniscript)** and — because the sketch uses a *relative* `older(N)` — **BIP-68/112 only** (not BIP-65, which is for an absolute `after()` /CLTV locktime) from **OUT** to **IN**, and add a miniscript satisfaction path to the signer — which is precisely why the design **selects 2-of-3** and rejects the timelock, keeping the whole wallet inside standard, Deployed multisig. Either way the policy must be fixed *before* the descriptor is frozen (not retrofittable). **Before any descriptor is called canonical, run a coordinator interop matrix** — exact versions of Sparrow, Bitcoin Core, Specter, Nunchuk, Electrum — with export/import tests for the descriptor checksum, key origins, partial signatures, and unknown/proprietary PSBT fields.

QK2 Standards Register · working draft v1 · 2026-08-17

Derivation path, descriptor, checksum, and addresses are candidate values from a pure-stdlib reference computation, cross-checked against the published BIP-32 and bech32 reference vectors and the secp256k1 root fingerprint `73c5da0a` (evidence level 2). Loading the example descriptor into Sparrow/ColdCard is an **interoperability objective pending repeatable tests**, not a confirmed result. Companion to the QuietKey QK2 Blueprint and the QKEC-1 Specification.